



SIMATS Online Programming Lab
C PROGRAMMING

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QUESTIONS 18 / 18 completed

Sum(ODD,EVEN)

SUM(ODD,EVEN)
C Program to Find the Sum of Even and Odd Numbers.
Problem Description
The program takes the number N and finds the sum of odd and even numbers from 1 to N.
Testcases:

PROGRAM EDITOR C Run Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int n,i,evensum=0,oddsum=0;
5     printf("enter the number :");
6     scanf("%d",&n);
7     for(i=1;i<=n;i++)
8     {
9         if(i%2==0)
10             evensum=evensum+i;
11         else
12             oddsum=oddsum+i;
13     }
14     printf("sum of even numbers=%d\n",evensum);
15     printf("sum of odd numbers=%d\n",oddsum);
16     return 0;
```

INPUT
10



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Natural

NATURAL
Write a program that prints the first n natural numbers using a for loop.

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```
1 #include<stdio.h>
2 int main()
3 {
4     int i;
5     for(i=1;i<=10;i++)
6     {
7         printf("%d\n",i);
8     }
9     return 0;
10 }
```

INPUT
Your INPUT goes here!



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Sum

SUM
Write a program that calculates the sum of the first n natural numbers using a while loop.

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```
1 #include<stdio.h>
2 int main ()
3 {
4     int n,i=1,sum=0;
5     printf("enter the number:\n");
6     scanf("%d",&n);
7     while(i<=n)
8     {
9         sum=sum+i;
10        i++;
11    }
12    printf("sum=%d",sum);
13    return 0;
14 }
```

INPUT
15



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Table

TABLE

Write a program that prints the multiplication table of a given number using a for loop.

PROGRAM EDITOR
C
Run Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int i,n;
5     printf("enter the number :");
6     scanf("%d",&n);
7     for (i=1;i<=10;i++)
8     {
9         printf("%d x %d=%d\n",n,i,n*i);
10    }
11    return 0;
12 }
```

INPUT
6



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Prime

PRIME

Write a program that takes a number as input and determines whether it is a prime number or not using a while loop.

PROGRAM EDITOR
C
Run Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int i,n;
5     scanf("%d",&n);
6     for (i=2;i<n;i++)
7     {
8         if(n%i==0)
9         {
10            printf("not prime");
11            return 0;
12        }
13        printf("prime");
14        return 0;
15    }
```

INPUT
3



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QUESTIONS 18 / 18 completed

Count Vows

COUNT VOWS

Write a program that takes a string as input and counts the number of vowels in the string using a for loop.

PROGRAM EDITOR
C
Run Save

```
1 #include<stdio.h>
2 int main()
3 {
4     char str[20];
5     int i,count=0;
6     printf("enter the string");
7     scanf("%19s",str);
8     for (i=0;str[i]!='\0';i++)
9     {
10        if (str[i]=='a' || str[i]=='e' || str[i]=='o' || str[i]=='u')
11            count++;
12    }
13    printf("number of vowels =%d", count);
14    return 0;
15 }
```

INPUT
apple



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QUESTIONS 18 / 18 completed

Fibonacci

FIBONACCI
Write a program that prints the Fibonacci series up to a given number using a while loop.

PROGRAM EDITOR

```
1 #include<stdio.h>
2 int main()
3 {
4     int n,i=1,a=0,b=1,c;
5     scanf("%d",&n);
6     while(i<=n)
7     {
8         printf("%d ",a);
9         c=a+b;
10        a=b;
11        b=c;
12        i++;
13    }
14    return 0;
15 }
```

INPUT
4

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Pyramid

PYRAMID
Write a program that prints a pyramid pattern of stars using nested loops.

PROGRAM EDITOR

```
1 #include<stdio.h>
2 int main()
3 {
4     int i,j,n;
5     scanf("%d",&n);
6     for(i=1;i<=n;i++)
7     {
8         for(j=1;j<=n-i;j++)
9             printf(" ");
10        for(j=1;j<=2*i-1;j++)
11            printf("*");
12        printf("\n");
13    }
14    return 0;
15 }
```

INPUT
5

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Diamond

DIAMOND
Write a program that prints a diamond pattern of stars using nested loops.

PROGRAM EDITOR

```
1 #include<stdio.h>
2 int main()
3 {
4     int n,i,j,isprime;
5     scanf("%d",&n);
6     if (n>=2)
7         printf("2");
8     for(j=3;j<=n;j++)
9     {
10        isprime=1;
11        for(i=2;i<j;i++)
12            if(j%i==0)
13            {
14                isprime=0;
15                break;
16            }
17    }
```

INPUT
10



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QUESTIONS 18 / 18 completed

R- triangle

R- TRIANGLE
Write a program that prints a right-angled triangle pattern of stars using nested loops.

PROGRAM EDITOR
C

```
1 //inverted right triangle
2 #include<stdio.h>
3 int main()
4 {
5     int i,j,n;
6     scanf("%d",&n);
7     for(i=n;i>=1;i--)
8     {
9         for(j=1;j<=i;j++)
10             printf("* ");
11         printf("\n");
12     }
13     return 0;
14 }
15
```

INPUT
5





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Hollow Square

HOLLOW SQUARE
Write a program that prints a hollow square pattern of stars using nested loops.

PROGRAM EDITOR
C

```
1 //square
2 #include<stdio.h>
3 int main()
4 {
5     int n,i,j;
6     scanf("%d",&n);
7     for(i=1;i<=n;i++)
8     {
9         for(j=1;j<=n;j++)
10             printf(" ");
11         printf("\n");
12     }
13     return 0;
14 }
```

INPUT
5





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Hollow Diamond

HOLLOW DIAMOND

Write a program that prints a hollow diamond pattern of stars using nested loops.

PROGRAM EDITOR

C

Run Save

```
1 //hollow pyramid
2 #include<stdio.h>
3 int main()
4 {
5     int i,j,n;
6     scanf("%d",&n);
7     for (i=1;i<=n;i++)
8     {
9         for (j=1;j<=n-i;j++)
10             printf(" ");
11         for (j=1;j<=2*i-1;j++)
12             if(i==1||i==n||j==1||j==2*i-1)
13                 printf("*");
14             else
15                 printf(" ");
16         printf("\n");
17     }
```

INPUT

5



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Pascal's triangle

PASCAL'S TRIANGLE

Write a program that prints a Pascal's triangle pattern using nested loops.

PROGRAM EDITOR

C

Run Save

```
1 //hollow diamond
2 #include<stdio.h>
3 int main()
4 {
5     int i,j,n;
6     scanf("%d",&n);
7     for(i=1;i<=n;i++)
8     {
9         for(j=1;j<=n-i;j++)
10             printf(" ");
11         for(j=1;j<=2*i-1;j++)
12             printf("*");
13         printf("\n");
14     }
15     for(i=n-1;i>=1;i--)
16     {
17         for(j=1;j<=n-i;j++)
18             printf(" ");
19         for(j=1;j<=2*i-1;j++)
20             printf("*");
21         printf("\n");
22     }
```

INPUT

5